Thin film CPA with nonlocal corrections

Corrections within the Feibelman d-parameter formalism

We start with the notes here: https://hackmd.io/@aligho/rJnJYnKixl (https://hackmd.io/@aligho/rJnJYnKixl) which derive the Feibelman d-parameters and their influence on the non-retarded reflection and transmission coefficients. Since the non-retarded surface plasmon dispersion is obtained as the pole of the reflection coefficient, we have:

$$(\varepsilon_d + \varepsilon_m) + k(\varepsilon_d - \varepsilon_m)(d_\perp - d_{||}) = 0$$

We take $\varepsilon_m = \varepsilon_\infty - \omega_p^2/\omega(\omega + i\gamma)$ in order to obtain an implicit equation for the frequency ω:

$$\Big[(\varepsilon_d + \epsilon_\infty)\omega(\omega + i\gamma) - \omega_p^2\Big] + k\Big[(\varepsilon_d - \varepsilon_\infty)\omega(\omega + i\gamma) + \omega_p^2\Big](d_\perp - d_{||}) \equiv \Big[\varepsilon_+\omega(\omega + i\gamma) - \omega_p^2\Big] + k\Big[\varepsilon_-\omega(\omega + i\gamma) + \omega_p^2\Big](d_\perp - d_{||}) = 0$$

This is simply a quadratic equation with $a = \varepsilon_+ + \varepsilon_- k(d_\perp - d_{||})$, $b = i\gamma\Big[\varepsilon_+ + \varepsilon_- k(d_\perp - d_{||})\Big]$ and, finally, $c = -\omega_p^2\Big[1 - k(d_\perp - d_{||})\Big]$. Putting all of this together, and taking the positive root, we obtain:

$$\frac{-i\gamma\Big[\varepsilon_+ + \varepsilon_- k(d_\perp - d_{||})\Big] + \sqrt{-\gamma^2\Big[\varepsilon_+ + \varepsilon_- k(d_\perp - d_{||})\Big]^2 + 4\omega_p^2\Big[1 - k(d_\perp - d_{||})\Big]\Big[\varepsilon_+ + \varepsilon_- k(d_\perp - d_{||})\Big]}}{2[\varepsilon_+ + \varepsilon_- k(d_\perp - d_{||})]}$$

Which we readily simplify to:

$$-\frac{i\gamma}{2} + \frac{1}{2}\sqrt{-\gamma^2 + 4\omega_p^2\frac{[1 - k(d_\perp - d_{||})]}{[\varepsilon_+ + \varepsilon_- k(d_\perp - d_{||})]}},$$

which is consistent with **equation 8.23** in Plasmonics and Light–Matter Interactions in Two-Dimensional Materials and in Metal Nanostructures (https://link.springer.com/book/10.1007/978-3-030-38291-9).

We use this expression to find the lowest order corrections to the surface plasmon dispersion due to having non-zero Feibelman d-parameters. For simplicity, we assume the Feibelman parameters are real valued. Note that in the absence of Feibelman corrections, and in the small scattering limit, we obtain:

$$\omega_{sp} = \frac{\omega_p}{\sqrt{\varepsilon_+}},$$

which is simply the familiar condition $\varepsilon_d + \varepsilon_m = 0$ (which gives the familiar $\omega_p/\sqrt{2}$ dispersion in the absence of ionic screening at a metal-vacuum interface. In the presence of d-parameter corrections, we have, instead,

$$\begin{aligned}\frac{\omega_p}{\sqrt{\varepsilon_+}}\sqrt{\frac{1-k(d_\perp - d_{||})}{1+(\varepsilon_-/\varepsilon_+)k(d_\perp - d_{||})}} &\approx \frac{\omega_p}{\sqrt{\varepsilon_+}}\left[1-\frac{1}{2}k(d_\perp - d_{||})\right]\left[1-\frac{1}{2}\frac{\varepsilon_-}{\varepsilon_+}k(d_\perp - d_{||})\right] \\ &\approx \omega_{sp}^0\left[1-\frac{\varepsilon_+ + \varepsilon_-}{2\varepsilon_+}k(d_\perp - d_{||})\right]\end{aligned}$$

For $\varepsilon_d = \varepsilon_\infty = 1$, we have $\varepsilon_+ + \varepsilon_- \equiv 2\varepsilon_d = 2$ and $\varepsilon_+ = 2$, which gives us:

$$\omega_{sp} \approx \omega_{sp}^0\left[1-\frac{1}{2}k(d_\perp - d_{||})\right],$$

which is **equation 8.24a** in Plasmonics and Light–Matter Interactions in Two-Dimensional Materials and in Metal Nanostructures (https://link.springer.com/book/10.1007/978-3-030-38291-9).

Corrections within the hydrodynamic approach

We already have a set of notes (https://hackmd.io/@aligho/HJdkkZS_xl (https://hackmd.io/@aligho/HJdkkZS_xl)) which derive, in detail, the hydrodynamic corrections to plasma oscillations in metal-insulator, metal-insulator-metal and insulator-metal-insulator configurations (the last configuration corresponds to the thin film).

In order to make direct comparisons with the Feibelman d-parameter approach, let's first examine the metal-insulator configuration in the non-retarded regime. In the notation of the aforementioned notes, in the non-retarded regime, we have $Q_1 = Q_2 = Q$, which gives for the plasma dispersion:

$$1 = -\frac{\varepsilon_m}{\varepsilon_d} + \frac{q}{Q'}\frac{\varepsilon_\infty - \varepsilon_m}{\varepsilon_\infty}$$

To make a direct comparison to the last equation of the previous section, we take $\varepsilon_\infty = 1$ and also use the fact that (in the absence of dissipation):

$$k_{nl}^2 \equiv \frac{\omega^2 - \omega_p^2}{\beta^2},$$

where $\beta^2 = 3v_F^2/5$ comes from the pressure of a three dimensional electron liquid at high frequencies. Note that near the surface plasmon frequency, $k_{nl}^2 < 0$ and $Q'^2 \equiv q^2 - k_{nl}^2 \approx -k_{nl}^2$. Putting everything together, we obtain the dispersion relation:

$$2 = \frac{\omega_p^2}{\omega^2} + \frac{q}{\sqrt{q^2 + (\omega_p^2 - \omega^2)/\beta^2}}\frac{\omega_p^2}{\omega^2} = \frac{\omega_p^2}{\omega^2}\left[1 + \frac{q\beta}{\sqrt{(q\beta)^2 + (\omega_p^2 - \omega^2)}}\right]$$

The solution to this implicit dispersion relation is given by **equation 6.23** in [Plasmonics and Light–Matter Interactions in Two-Dimensional Materials and in Metal Nanostructures (https://link.springer.com/book/10.1007/978-3-030-38291-9)]

$$\omega^2 = \frac{1}{2}\left[\omega_p^2 + \beta^2 q^2 + \beta q\sqrt{2\omega_p^2 + \beta^2 q^2}\right] = \frac{1}{4}\left[\beta q + \sqrt{2\omega_p^2 + \beta^2 q^2}\right]^2 \rightarrow$$

$$(q\beta)^2 + \omega_p^2 - \omega^2 = \frac{1}{2}\left[\omega_p^2 + \beta^2 q^2 - \beta q\sqrt{2\omega_p^2 + \beta^2 q^2}\right] = \frac{1}{4}\left[\beta q - \sqrt{2\omega_p^2 + \beta^2 q^2}\right]^2$$

$$\rightarrow \sqrt{(q\beta)^2 + \omega_p^2 - \omega^2} = \frac{1}{2}\left[\sqrt{2\omega_p^2 + \beta^2 q^2} - \beta q\right]$$

We now check that this satisfies the dispersion:

$$2 = \frac{4\omega_p^2}{\left[\beta q + \sqrt{2\omega_p^2 + \beta^2 q^2}\right]^2}\left[1 - \frac{2q\beta}{\beta q - \sqrt{2\omega_p^2 + \beta^2 q^2}}\right] =$$

$$\frac{4\omega_p^2}{\left[\beta q + \sqrt{2\omega_p^2 + \beta^2 q^2}\right]}\left[\frac{-q\beta - \sqrt{2\omega_p^2 + \beta^2 q^2}}{-2\omega_p^2}\right] = 2,$$

QED.

Obtaining the CPA condition from the surface plasmon dispersion

See https://hackmd.io/@aligho/HJdkkZS_xl (https://hackmd.io/@aligho/HJdkkZS_xl) and https://hackmd.io/@aligho/SymTGLcPxg (https://hackmd.io/@aligho/SymTGLcPxg)
$\tanh(Q_2 w/2) = -\varepsilon_m Q_1/\varepsilon_d Q_2$

We take $\varepsilon_d = 1$ and we take $q = 0 \rightarrow Q_2 = in\omega/c, Q_1 = i\omega/c$

Therefore, we have (replacing w with d and ω/c with k to be consistent with the cpa notation):

$$\tanh(inkd/2) = -n \rightarrow \frac{e^{inkd/2} - e^{-ink\omega d/2}}{e^{inkd/2} + e^{-inkd/2}} = -n$$

Taking $y = e^{inkd}$, we obtain:

$$\frac{1-y}{1+y} = n \rightarrow n(1+y) = 1-y \rightarrow y = \frac{1-n}{1+n}$$

For the other mode, we obtain:

$$\frac{e^{inkd/2} + e^{-inkd/2}}{e^{inkd/2} - e^{-inkd/2}} = -n \rightarrow (y+1) = n(1-y) \rightarrow y = \frac{n-1}{1+n}$$

For a monolayer

The monolayer plasmon dispersion is given by:

$$1 + \frac{i\sigma(q,\omega)\sqrt{q^2 - \omega^2/c^2}}{2\omega\varepsilon_0} = 0 \rightarrow 1 - \frac{\sigma(0,\omega)}{2c\varepsilon_0} = 0 \rightarrow \sigma(\omega) = 2c\varepsilon_0,$$

which is the same as the relation found in https://hackmd.io/@aligho/HyMyvezclx (https://hackmd.io/@aligho/HyMyvezclx)